Table 1: Client-Side Encryption Complexity Per Round

Scheme	Packs enc.	Cost / pack	Encryption complexity
FedAvg+CKKS	J	$\mathcal{O}(n \log n)$	$\Theta(d \log n)$
FedPHE	sJ	$\mathcal{O}(n \log n)$	$\Theta(sd \log n)$
ESAFL	$\lceil J/2 \rceil$	$\mathcal{O}(n \log n)$	$\Theta(d \log n)$
MASER	ζJ	$\mathcal{O}(n \log n)$	$\Theta(\zeta d \log n)$
SAMK	$J + N$	$\mathcal{O}(n \log n) + \mathcal{O}(\log^3 \mathcal{N})$	$\Theta(d \log n) + \Theta(N \log^3 \mathcal{N})$
ECPFL (ours)	$1{+}k$	$\mathcal{O}(n \log n)$	$\Theta\big((k{+}1)\, n \log n\big) = o(d \log n)$

Table 2: Per-Round Structure, Dropout Robustness and Setup Assumptions

Scheme	Comm. rounds	Client uplink (per round)	Dropout requirement	Trusted setup
FedAvg+CKKS	1	$\Theta(d \log Q)$	any subset	shared sk
FedPHE	2 (sketch $\rightarrow$ ct)	$\Theta(k_s + sd \log Q)$	any subset of $S^{(t)}$	KDC every τ rounds
ESAFL	1	$\Theta(d \log Q)$	**all N clients**	trusted AS (one-time)
MASER	2 (vote $\rightarrow$ ct) + shares	$\Theta(d) + \Theta(\zeta J n \log Q)$ (ct & share)	**all N clients**	trusted key manager
SAMK	1	$\Theta\big(d \log Q + \frac{Nd}{n} \log \mathcal{N}\big)$	**any non-empty subset**	**none**
ECPFL (ours)	1	$\Theta((k{+}1) n \log Q)$	any $\widehat{S}^{(t)}$ with $\lvert\widehat{S}^{(t)}\rvert \geq \tau$	offline KGC (one-time shares)

$J = \lceil d/n \rceil$; ζ: MASER keep-ratio; k_s: FedPHE sketch dimension; N: clients (SAMK interpolation points); $\mathcal{N}$: Paillier modulus. MASER uploads plaintext masks in an extra voting round; its partial-decryption shares are ciphertext-sized, hence counted twice.